

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2023

Bachelor in Computer Applications

Course Title: **Mathematics II**

Code No: **CACS154** Semester: **II**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. Evaluate the limit $\lim_{x \rightarrow \theta} \frac{x \cos \theta - \theta \cos x}{x - \theta}$.
3. Find the derivative of $y = e^{3x - 1}$ by the definition method.
4. Verify Rolle's theorem for the function $f(x) = \sin x, x \in [0, \pi]$ and find the point on the curve where the tangent is parallel to the x-axis.
5. Find maximum and minimum values of the function $f(x) = x^3 + 6x^2 + 9x - 2$. Also find the point of inflection, if any.
6. Evaluate the integrals.
a. $\int \frac{2x + 5}{\sqrt{x^2 + 5x}} dx$ b. $\int e^{2x} \log x dx$
7. Evaluate $\int_0^2 \sqrt{1 + x^3} dx$ by using Simpson's $\frac{1}{3}$ rule, taking $n = 4$.
8. Solve the differential equation: $2 \frac{dy}{dx} = \frac{2y}{x} + \frac{y^2}{x^2}$.

Group C

Attempt any TWO questions [2x10=20]

9. Define pivot element, pivot column, and consistency in a system of equations. Using the simplex method, maximize $F = 5x - 3y$ subject to $3x + 2y \leq 6, -x + 3y \leq -4, x \geq 0, y \geq 0$.
10. a. Verify Lagrange's mean value theorem for the function $f(x) = \sqrt{x - 1}, x \in [1, 3]$.
b. Solve the differential equation $x \frac{dy}{dx} = x^2 + y^2$.
11. Compute the approximate value of the integral $\int \frac{1}{1 + x^2} dx$ by using the composite trapezoidal rule with three points, and compare the result with the actual value. Determine the error formula and numerically verify an upper bound on it.